

Redis – High Performance In-Memory Data Store

Prepared By: Dr. D. P. Singh

What is Redis?

- Open-source **in-memory key-value** store
- Used for **caching, messaging, and real-time systems**
- Works on RAM → extremely low latency

Why Redis?

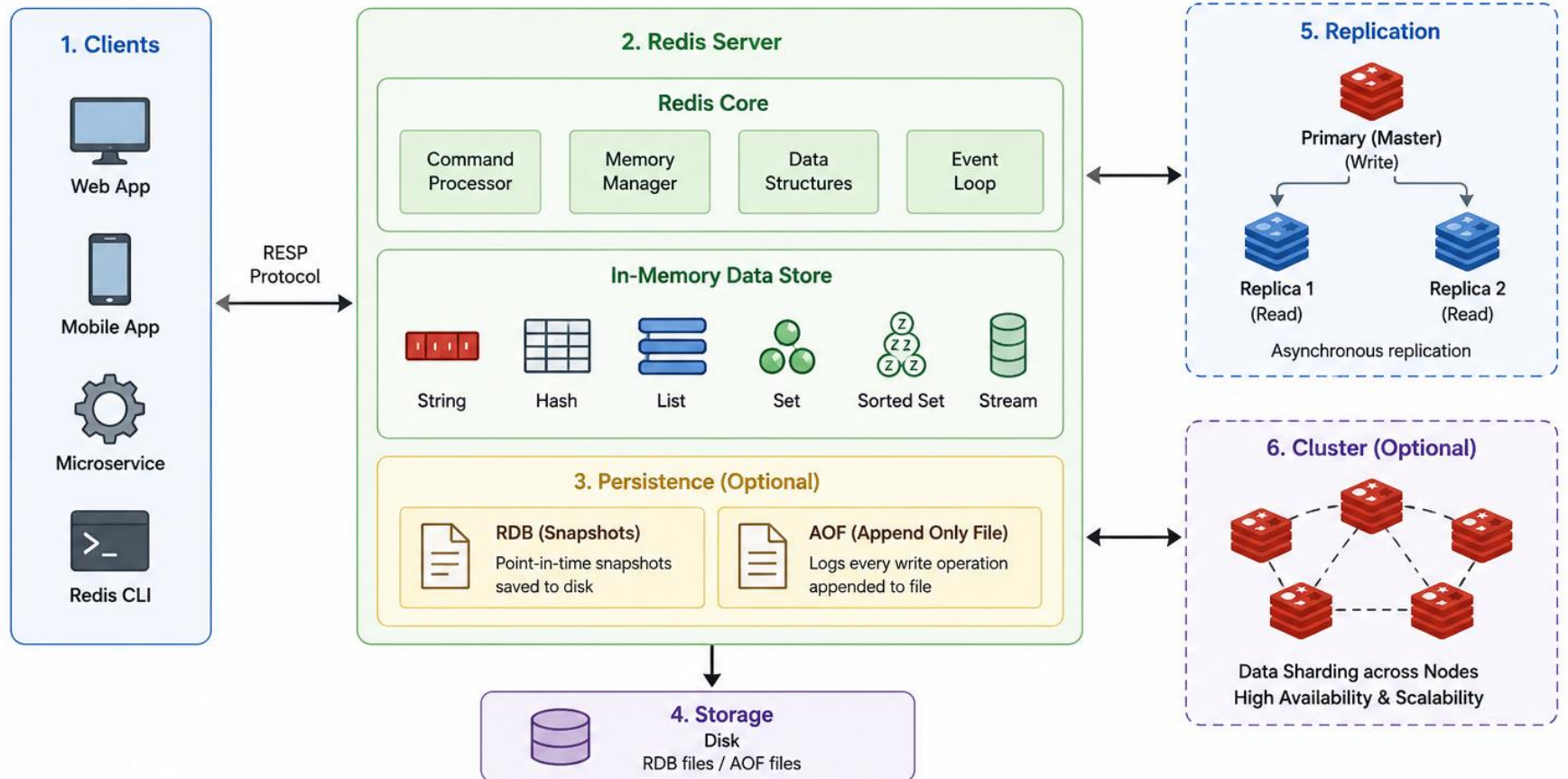
- **Faster** than disk-based DBs
- Handles millions of **requests/sec**
- Used in real-time systems

Redis Architecture

- Client → Redis Server → (Optional) Database
- Redis runs as a background service (daemon)
- Uses TCP protocol (default **port 6379**)
- Stores data in memory + optional persistence
- **RESP (Redis Serialization Protocol)** is the application-layer protocol that clients use to communicate with Redis. It defines how commands and responses are encoded over a TCP connection.

Redis Architecture

Redis Architecture



1. Clients

Applications connect to Redis using RESP protocol

2. Redis Server

Handles client requests, manages memory and executes commands

3. Persistence

Data can be persisted using RDB snapshots and/or AOF logs

4. Storage

Snapshot or log files are stored on disk for durability

5. Replication

Data is replicated from master to replicas for redundancy & read scaling

6. Cluster

Data is sharded across multiple nodes for horizontal scaling and high availability

Installation & Setup

- Installation
 - `sudo apt get install redis-server`
- Start Redis
 - `redis-server`
- Test Redis
 - `redis-cli ping` → PONG

Basic Commands

- **Set and Get:**
 - SET name "Devesh"
 - GET name
 - **Output:**
 - "Devesh"
- **Delete Key:**
 - DEL name
- **Check existence:**
 - EXISTS name

Data Structures

- Strings, Hashes, Lists, Sets, Sorted Sets, Streams
- **Strings:**
 - SET user:1 "Ramesh"
- **Hash (like JSON object):**
 - HSET user:1 name "Ramesh" age 21
 - HGETALL user:1
- **List:**
 - LPUSH queue "task1"
- **Set:**
 - SADD tags "redis" "db" "cache"
 - SMEMBERS tags

Redis in Industry (Real Time Systems)

- Instagram → feed caching
- Twitter/X → timeline cache
- Uber → real-time location updates
- Amazon → session + product caching

Advantages

- Very fast
- Scalable
- Supports multiple data structures

Limitations

- Memory expensive
- Not for large storage
- Risk of data loss

- Thank You